PART 5

Medical Emergencies

IN THIS PART

1. Describe an approach to a patient with altered mental status and the reversible causes to check, such as low blood glucose.
2. Recognize stroke with a validated stroke screen and state why time matters.
3. Describe the response to a seizure.
4. Recognize sepsis and state first priorities.
5. Describe first priorities in poisoning and overdose.
6. Recognize acute abdominal emergencies and gastrointestinal bleeding.

Not every emergency begins with an injury or a collapse. Many begin with a change that is easily explained away: a patient who is a little confused, a limb that feels odd, a fever that will not settle, a stomach ache that has lasted too long. The skill of this Part is to take these small changes seriously and to know which of them can wait and which cannot.

The Part follows the problems you are most likely to meet. Entry 5.1 gives a way to approach any patient whose thinking or alertness has changed and starts with the cause that can be checked and reversed in minutes, low blood glucose. Entry 5.2 covers stroke, where time decides how much of the brain can be saved.

Entry 5.3 covers seizures, 5.4 covers sepsis, 5.5 covers poisoning and overdose, and 5.6 covers abdominal pain and bleeding from the gut.

These entries rely on Part 1 more than the others do. The primary survey, vital signs and history are how you find these problems, and the same order applies. Your scope of practice and local protocol decide what you may do, and the medicines used in these conditions are given by trained clinicians under medical direction.

5.1 Altered Mental Status and Hypoglycemia

A change in how a person thinks, speaks or responds is one of the most important signs in emergency care, and it has many causes. The aim is to check the causes that kill quickly and can be reversed, and to call for help while you do.

WHAT IT IS. Altered mental status means that a patient is less alert, less oriented or behaving differently than usual. It runs from mild confusion to unresponsiveness. Entry 1.3 gave you the AVPU scale and the Glasgow Coma Scale to describe it, and the trend over time is important in showing when new action is needed. A patient who was alert 10 minutes ago and is drowsy now is worsening, whatever the cause.

WHY IT MATTERS. The brain is sensitive to its supply of oxygen and glucose and to what is in the blood. A fall in either, an infection, a poison, a head injury or a stroke can change behavior long before other signs appear. Patients who are confused may be unable to give a history, may refuse help, or may be mistaken for intoxicated. Do not assume that confusion is caused by alcohol or by old age until the reversible causes have been excluded.

REVERSIBLE CAUSES. Work through a short list: the airway and oxygen supply, the blood glucose, signs of stroke, head injury, infection, poisoning or a recent seizure, extreme heat or cold, and low blood pressure. Each of these is covered in this book: oxygen in Part 2, shock in Entry 3.5, head injury in Entry 4.3, heat and cold in Entry 4.6, and the others in this Part.

Figure 19 sets out the approach as a flow; notice that airway, breathing and glucose come before the search for other causes.

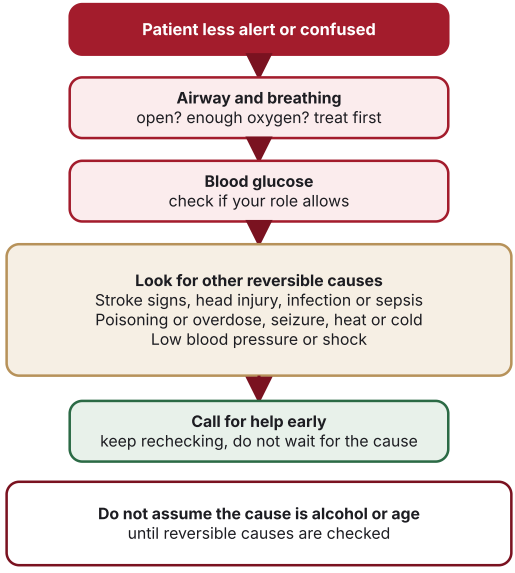


Figure 19. An approach to a patient with altered mental status, from airway and breathing to the reversible causes.

TALKING WITH A CONFUSED PATIENT. Introduce yourself by name, speak slowly and in short sentences, and ask one question at a time. Ask the patient's name, where they are and what day it is, and note the answers. Do not argue with what a confused person says. Ask family, carers or bystanders what the patient is usu-

ally like and when the change began, because the answer tells you whether this is new. A medical alert bracelet or card may give the diagnosis and the medicines.

ALCOHOL AND OTHER SUBSTANCES. A patient who smells of alcohol can still have low blood glucose, a head injury, a stroke or a serious infection, and drowsiness in an intoxicated patient can hide all of them. Treat altered mental status in a patient who has been drinking as a medical emergency, check the same causes, and keep the patient on their side if drowsy, because vomiting is a risk.

LOW BLOOD GLUCOSE. Hypoglycemia is low blood glucose, and it is the cause that is quickest to check and quickest to reverse. Entry 1.3 gave the usual low reading, 70 milligrams per deciliter (mg/dL) or 3.9 millimoles per liter, which differs from person to person. It most often occurs in people who take certain diabetes medicines that lower blood glucose, and it can follow missed meals, extra exercise or alcohol. Early signs include shakiness, sweating, hunger, headache and anxiety. Later signs include confusion, odd behavior, slurred speech, drowsiness, seizures and loss of consciousness. A high reading can also cause drowsiness: a person with diabetes who is very unwell with deep, fast breathing, thirst and vomiting may have diabetic ketoacidosis, so report any high reading and call for help. Slow breathing, very small pupils and unresponsiveness suggest an opioid overdose, which Entry 5.5 covers.

PROCEDURE: HELP A CONSCIOUS PATIENT WITH LOW BLOOD GLUCOSE

Purpose: to restore the patient's blood glucose with a fast-acting sugar. Scope: follows your training level and local protocol; the product and the amount are set by your protocol and medical direction.

1. Make the scene safe and carry out the primary survey (Entry 1.3).
2. Check the blood glucose with a meter if your role allows it.
3. Confirm that the patient is awake, can sit up, and can swallow safely.
4. Give a fast-acting sugar that your protocol names, as it directs.
5. Keep the patient seated and calm, and stay with them.
6. Recheck the blood glucose after about 15 minutes, and repeat the treatment if it is still low, as your protocol directs.
7. When the patient has recovered, tell the responders what happened and when, because the cause still needs to be found.

> **⚠ SAFETY**
>
> Give nothing by mouth to a patient who is drowsy, confused to the point of not following commands, has a seizure, or cannot swallow safely. Call for help. Place a patient who is unconscious but breathing in the recovery position (Entry 2.2) unless a spinal injury is suspected, and stay with the patient. Giving medicines is outside many roles; follow your scope and medical direction.

STOP AND CALL FOR HELP IF the patient cannot swallow, becomes drowsier or has a seizure, does not improve after treatment, or the blood glucose cannot be measured and the patient is worsening.

The steps work in this order because swallowing is the safety check. A patient who is awake and swallowing normally can be helped by mouth, and a patient who is not can choke on a drink.

Rechecking the reading tells you whether the treatment worked and avoids both a missed relapse and an unnecessary second portion of sugar.

RED FLAGS

A patient who is hard to wake or cannot follow commands, new confusion with a very fast or slow pulse, a seizure, signs of stroke, a head injury, slow breathing, or a low glucose reading that does not improve. Call emergency services at once.

NURSING AND RESPONDER PRIORITIES. Check airway, breathing and glucose early, and call for help for any patient whose cause is not clear.

See also: Entry 1.3 (primary survey and glucose check); Entry 2.2 (recovery position); Entry 5.2 (stroke screen); Entry 6.1 (low glucose in children).

5.2 Stroke

Stroke is damage to part of the brain from a blocked or burst blood vessel. It is a time-critical emergency. The treatments that limit damage work best when they are started early, so recognizing a stroke quickly and calling for help is the most useful thing a bystander can do.

WHAT IT IS. Most strokes occur when a clot blocks an artery that supplies the brain. The brain tissue beyond the block is starved of oxygen and begins to die. A smaller number occur when a vessel bursts and bleeds into the brain. A transient ischemic attack, sometimes called a mini-stroke, is a brief blockage that clears on its own. It is a warning and needs urgent assessment because a full stroke may follow.

Figure 20 shows a simplified side view of the brain; notice that the front, middle and rear regions are supplied mainly by different arteries, which is why different strokes cause different signs.

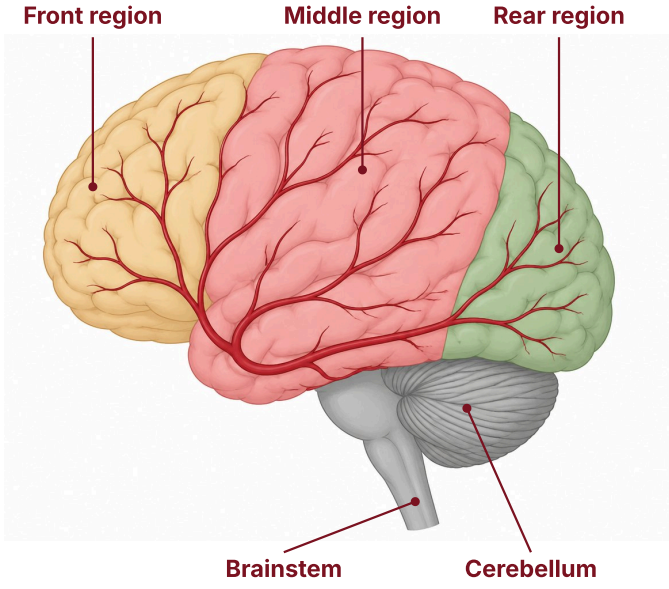


Figure 20. A simplified side view of the brain showing the front, middle and rear regions, each supplied mainly by a different main artery.

RECOGNITION. The signs begin suddenly. The face droops on one side. An arm or leg becomes weak or numb, usually on one side. Speech is slurred or confused, or the patient cannot find words or understand. Other signs include sudden loss of vision, double vision, sudden severe headache, dizziness, and loss of balance. A short screening tool checks the face, arms and speech and asks the time. One widely taught version is FAST, for Face, Arm, Speech and Time. Some versions add Balance and Eyes, because a stroke in the back of the brain can cause sudden dizziness, loss of balance or loss of vision without a drooping face. Ask the patient

to smile, to raise both arms and hold them, and to repeat a short sentence. If any of these is abnormal, treat the patient as having a stroke and note the time.

Figure 21 shows the signs on a face and on the arms; notice the droop on one side of the face and the arm that drifts lower.

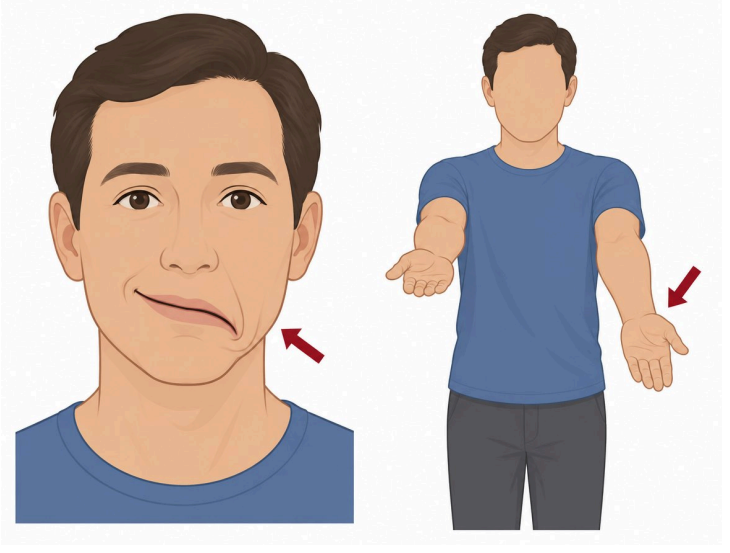


Figure 21. Facial droop and arm weakness on one side, two signs of the stroke screen.

TWO KINDS OF STROKE. A clot and a bleed need opposite treatments, which is why a scan must come first and why you do not give a medicine that affects clotting on your own judgment. You cannot tell the two apart at the bedside. A bleed more often starts with a sudden severe headache, vomiting or a rapid fall in alertness, but a clot can look the same. Both need the same response from you: call, protect the airway and write down the times.

CONDITIONS THAT LOOK LIKE STROKE. Low blood glucose, seizures with weakness afterward, migraine and some infections can mimic a stroke. You do not need to decide which is which. Treat sudden one-sided weakness, facial droop or speech trouble as a stroke until a clinician shows otherwise, and check the blood glucose while you wait.

AFTER A BRIEF EPISODE. Symptoms that appear and then go away within minutes or hours may be a transient ischemic attack. The patient may feel well and refuse help. Explain that this is a warning and that the risk of a full stroke is highest in the first hours and days afterward. Treat it as an emergency and call emergency services even though the symptoms have gone.

> **REMEMBER**
>
> FAST: Face drooping, Arm weakness, Speech trouble, Time. The T is a reminder to call at once and to record when the symptoms began or when the patient was last known to be well.

WHY TIME MATTERS. The longer the brain goes without blood, the more of it dies, and the available treatments have time limits that are measured in hours. The most useful fact you can give the receiving team is the last time anyone saw the patient well. This may be earlier than the time the symptoms were noticed or the patient was found. A patient who wakes with symptoms was last known well at bedtime, not at waking, and the same applies to someone found on the floor: count from the last time anyone saw them well, and ask family, neighbors or carers when that was. Write down how you know, because the receiving team will ask how the time was worked out.

RED FLAGS

Sudden facial droop, arm or leg weakness, trouble speaking or understanding, sudden loss of vision, sudden severe headache, or loss of balance. Call emergency services at once and note the time the patient was last known to be well.

FIRST PRIORITIES. Call emergency services without delay, and say that you suspect a stroke. Note the time the symptoms began or the patient was last well. Check airway, breathing and circulation and the blood glucose, because low glucose can look like a stroke, but do not let the glucose check delay the call. Let the patient rest in the position of greatest comfort, or on their side if drowsy. Give nothing by mouth, because swallowing may be impaired. Do not give aspirin or any medicine unless medical direction tells you to, since some strokes are caused by bleeding. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Screen with a stroke tool, call early, and hand over the last known well time.

See also: Entry 1.3 (primary survey); Entry 5.1 (altered mental status); Entry 6.4 (older adults).

5.3 Seizures

A seizure is a burst of abnormal electrical activity in the brain. It can look dramatic, and the most useful thing a responder can do is also the simplest: stay calm, protect the patient from injury, and watch the clock.

WHAT IT IS. Seizures take many forms. A generalized tonic-clonic seizure causes loss of consciousness, stiffening and rhythmic jerking of the limbs, and is followed by a period of drowsiness

and confusion. Other seizures cause staring, brief lapses, or movements of part of the body. Seizures have many causes, including epilepsy, low blood glucose, head injury, fever in young children, alcohol withdrawal, poisoning and stroke.

FIRST AID DURING A SEIZURE. Note the time it begins. Ease the patient to the floor if they are standing, move hard objects away, and cushion the head. Loosen tight clothing at the neck. Turn them onto their side once the jerking stops, or sooner if they vomit, and keep the airway open. Entry 2.2 described the recovery position.

> ⚠ **SAFETY**
>
> Do not restrain the patient. Do not put anything in the mouth. Do not give food or drink until the patient is fully alert. Stay with them until they recover.

LONG AND REPEATED SEIZURES. A seizure that lasts longer than 5 minutes is a medical emergency, as is a series of seizures with no recovery between them. Treatment is given by clinicians. Your role is to protect the patient, keep the airway open as well as you can, and tell the dispatcher how long the seizure has lasted. Look for a medical alert bracelet or card and ask whether the patient has a seizure plan.

FEVER AND SEIZURES IN CHILDREN. Young children can have a seizure when they have a fever. It is frightening and usually brief, but the first one needs assessment by a clinician, and Entry 6.1 returns to children.

WHAT TO WATCH. Notice what the clinicians cannot: which part of the body moved first, whether the eyes turned to one side, how long it lasted, whether the patient answered during it, and how long the drowsiness lasted. Not every seizure convulses. If a

patient stares, makes chewing movements or wanders, stay close, guide them away from hazards, do not restrain them, and speak calmly. Tell the dispatcher what you saw.

AFTER THE SEIZURE. The patient will be drowsy and confused, and may not remember. Check breathing, speak calmly, and tell them where they are. Look for injuries. Check the blood glucose if your role allows.

RED FLAGS

A seizure that lasts longer than 5 minutes, repeated seizures without recovery between them, difficulty breathing afterward, an injury during the seizure, a first seizure, a seizure in water, a seizure in pregnancy, or a seizure in a patient with diabetes. Call emergency services at once.

NURSING AND RESPONDER PRIORITIES. Note the time, protect the patient, and call for help for any red flag.

See also: Entry 2.2 (recovery position); Entry 5.1 (altered mental status); Entry 6.1 (fever seizures in children).

5.4 Sepsis

Sepsis is the body's life-threatening response to an infection. The infection may be in the lungs, the urinary tract, the skin or the abdomen, and the response injures the organs. Sepsis can progress within hours, and the earlier it is recognized, the sooner treatment can start.

WHAT IT IS. Sepsis occurs when the body's response to an infection goes out of control and begins to damage its own organs. When it causes dangerously low blood pressure that does not recover, it is called septic shock. Anyone can develop sepsis, and risk

is higher in older adults, infants, people with weakened immune systems, people with long-term illness, and people with recent surgery or wounds.

RECOGNITION. Suspect sepsis in a patient who has or may have an infection and who also has any of these signs: a high or a low temperature, a fast heart rate, a fast breathing rate, new confusion or drowsiness, a low blood pressure, cool or mottled skin, or a patient who looks very unwell. In older adults a high temperature is less common and confusion may be the main sign; in infants, poor feeding, unusual sleepiness or a low temperature may be the signs. Early warning scores, which combine several vital signs into a single number, are used in many settings to flag deterioration, and your service will teach the one it uses.

Figure 22 shows the pathway from suspicion to action; notice that any one warning sign is enough to reach the call for help.

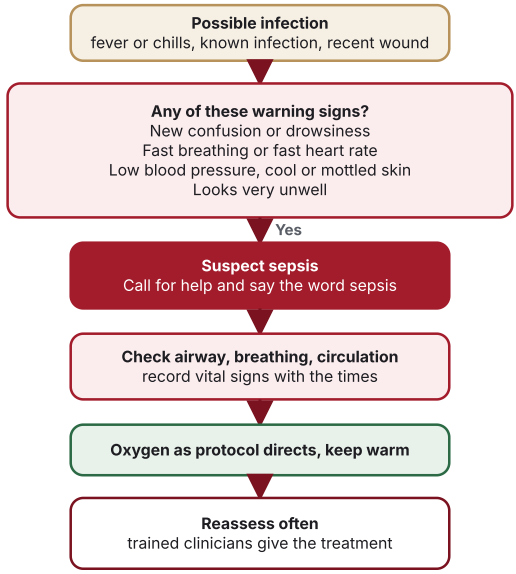


Figure 22. A sepsis recognition pathway from a suspected infection to a call for help.

RED FLAGS

A suspected infection with new confusion, a fast breathing rate, a fast heart rate, a low blood pressure, cool or mottled skin, or a patient who looks very unwell. Call for help at once and say that you suspect sepsis.

A patient who has had an infection and then becomes more unwell rather than better should be assessed again.

WHY SCREENING USES SEVERAL SIGNS. No single sign proves sepsis, and a normal temperature does not exclude it. A fast heart rate, a fast breathing rate or a fever each occur in many

harmless illnesses, so services screen for a combination and for change, often by adding the signs into an early warning score. A score is a prompt to call for help and is not a diagnosis. Entry 1.4 gave typical resting values; here the question is whether the numbers are moving in the wrong direction between sets.

SEPTIC SHOCK. Entry 3.5 explained shock from widened vessels. In sepsis the vessels both widen and leak, so the circulation is short of volume although no blood has been lost, and the blood pressure falls. Early on the skin may be warm and flushed; later it turns cool and mottled. A patient who is cool, mottled and newly confused is deteriorating.

HANDOVER. Entry 1.5 set out the structured handover. For sepsis, make sure it includes the suspected source of infection, the vital signs with their times, any new confusion, and the word sepsis.

FIRST PRIORITIES. Call for help early and say the word sepsis. Check airway, breathing and circulation, and take a full set of vital signs with the times. Give oxygen as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction. Treatment is given by trained clinicians under medical direction, so your part is to recognize the problem, record the times and bring the team.

NURSING AND RESPONDER PRIORITIES. Suspect sepsis in any unwell patient with a possible infection, check and record vital signs with times, call for help early, and reassess often.

See also: Entry 1.4 (vital signs); Entry 3.5 (shock); Entry 6.1 (children); Entry 6.4 (older adults).

5.5 Poisoning and Overdose

A poison is any substance that harms the body when it is swallowed, breathed in, injected or absorbed through the skin. Poisonings include medicines taken in the wrong amount or by the wrong person, household products, plants, alcohol, drugs of misuse, carbon monoxide and chemicals. Many are accidental, and some are intentional, which Entry 6.5 covers from the mental health side.

WHAT IT IS. Overdose is a poisoning by a medicine or a drug. The effects depend on the substance, the amount, the time and the person. Common patterns include drowsiness and slow breathing, agitation and a racing heart, vomiting and abdominal pain, seizures and confusion. Signs of some poisons take time to develop, so a patient who looks well may still be in danger.

WHY IT MATTERS. The history is the most useful tool. The substance, the amount, the time and the person's weight and age guide the experts who advise on treatment. Poisoning can also look like a stroke, a seizure or a head injury, and the primary survey and the blood glucose check will find the problems that need action right away.

FIRST PRIORITIES. Make the scene safe. The substance may harm you too, as carbon monoxide or a chemical spill can. Remove the patient from exposure if it is safe to do so. Check airway, breathing and circulation. Call emergency services and, in the United States, the national Poison Help line at 1-800-222-1222, which is staffed by experts who give advice for any poisoning. Collect the container, the package or a sample of what the patient took, and bring it. Do not make the patient vomit and do not give anything to drink unless the experts direct you.

COMMON POISONINGS. Household products, medicines left within reach of children, plants, alcohol and drug use account for many poisonings. Carbon monoxide, a gas without smell or color from faulty heaters and engines, causes headache, dizziness and confusion, and everyone in the same space may be affected, including pets. If it is safe, get everyone into fresh air and call emergency services; Entry 1.4 noted that a pulse oximeter can read normally in this poisoning, so a normal reading does not rule it out. Alcohol poisoning causes drowsiness, vomiting and slow breathing, and the patient should be kept on their side.

OPIOID OVERDOSE. Opioids are a class of drugs that include some pain medicines and illegal drugs. An overdose causes slow or absent breathing, unresponsiveness and very small pupils. Check for a response and for breathing, call emergency services, and give breaths or compressions if the patient is not breathing normally, as Entries 2.3 and 3.4 describe. Naloxone is a medicine that reverses the effects of an opioid overdose. Where naloxone is available and you are trained and permitted to use it, use it as the device directs, and stay with the patient, because the effects of naloxone can wear off before the opioid does and the patient can stop breathing again. Giving medicines is outside many roles; follow your scope and medical direction.

> ⚠ **SAFETY**
>
> Protect yourself from exposure and from needles. Wear gloves. If the scene is unsafe, such as a closed space that may hold poisonous gas, stay out and call for help. Do not give anything by mouth to a patient who is drowsy.

RED FLAGS

Drowsiness or slow breathing, a seizure, chest pain, a very fast or slow pulse, severe vomiting, burns around the mouth, or any poisoning in a child or an older adult. Call emergency services at once, and call the Poison Help line.

NURSING AND RESPONDER PRIORITIES. Protect yourself, call early, gather the container, and support airway and breathing.

See also: Entry 1.3 (primary survey); Entry 5.1 (altered mental status); Entry 6.1 (children); Entry 6.5 (self-harm).

5.6 Acute Abdominal Pain and Gastrointestinal Bleeding

Abdominal pain has dozens of causes, and most are mild. A few are emergencies, and they can be hard to tell apart from the others at the bedside. The safest rule is to look for features that make the pain dangerous.

WHAT IT IS. Acute abdominal pain is new pain in the belly. Causes include a bowel infection, appendicitis, gallstones, a blocked or perforated bowel, an ulcer, and bleeding from a vessel. Gastrointestinal bleeding is bleeding from the digestive tract. Bleeding from the upper gut, in the stomach or food pipe, can cause vomiting of blood, which may be bright red or look like coffee grounds. Bleeding from the lower gut usually causes bright red or maroon blood in the stool, while black, tarry, foul-smelling stools usually point to bleeding higher up in the gut.

RECOGNITION. Ask where the pain began, where it is now, how severe it is, and whether it has changed. Ask about vomiting, bowel changes, fever, recent injury, pregnancy, and medicines. Look at the patient's color and look at the abdomen without pressing hard. A

rigid, board-like abdomen, pain that gets worse with any movement, or a patient who curls up and will not let anyone touch the belly suggests an irritated lining, which is an emergency. Signs of shock with abdominal pain suggest bleeding.

RED FLAGS

Severe or worsening pain, a rigid abdomen, vomiting blood, black tarry stools, fainting, a fast pulse with cool pale skin, pain after an injury, or pain in a pregnant patient or an older adult. Call for help at once.

APPENDICITIS AND OTHER SURGICAL CAUSES. Appendicitis often begins as vague pain near the navel that moves to the lower right of the abdomen over hours, with loss of appetite, nausea and low fever. The pain worsens with movement, coughing or sneezing. Other emergencies that need surgery include a blocked or perforated bowel, a leaking blood vessel and an inflamed gallbladder. In any patient who could be pregnant, abdominal pain with a missed period or vaginal bleeding may mean an ectopic pregnancy, which can bleed heavily, and needs immediate help.

FIRST PRIORITIES. Check airway, breathing and circulation. Let the patient find the position of greatest comfort, and keep them lying down with the legs raised if they are faint or pale. Give nothing by mouth, because the patient may need surgery or may vomit. Take vital signs with the times, and watch for shock. Wear gloves, record what the patient has vomited or passed, and keep a sample if the clinicians ask. Do not give painkillers or medicines for the stomach unless medical direction directs. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Look for the dangerous features, give nothing by mouth, support the position of comfort, watch for shock, and call for help early.

See also: Entry 3.5 (shock); Entry 4.1 (abdominal injury); Entry 6.3 (pregnancy); Entry 6.4 (older adults).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Grace, a nursing student on a home-care placement, arrived for a visit and found Walter, an older man who lived alone, sitting in his kitchen. His daughter had told the agency that he seemed more confused than usual that morning. He did not know what day it was, and his speech was slow. Grace checked the room for hazards and put on gloves.

She worked through the survey she had practiced. His airway was open and his breathing was a little fast. His skin was cool and clammy. He was alert enough to say his name, but not to answer other questions. His face and arms looked symmetrical and he could squeeze her hands equally, so she saw no sign of a stroke. She phoned her supervisor, who asked for an ambulance to be sent and told her to check his blood glucose. It was low. She asked his daughter whether he took medicine for diabetes; the daughter said he did and that he had eaten nothing that morning. After checking that he was awake and able to swallow safely, she was told to follow the protocol for a conscious patient. Grace helped him take the sugar as directed, stayed with him, and rechecked the reading after about 15 minutes. He was clearer, and his speech improved.

Grace noted the time and the readings and left the ambulance call in place, because she did not know why his glucose had dropped. The paramedics arrived, and she handed over her findings and the times.

Care Priorities

CARE PRIORITIES

In medical emergencies the cause is usually unknown at the bedside, so the priorities have to work before the diagnosis does. First, protect the airway and breathing whatever the cause, because a patient who is drowsy and unexplained can still be kept safe. Second, look for the cause that is quick to check and can be reversed, and do that before the causes that only a hospital can sort out. Third, treat the change over time as more important than any single reading: write each observation with its time, because the trend is the part the receiving team cannot see for themselves. Give weight to what family or carers say is different from usual, since a confused patient cannot report their own baseline. When the story changes, start again from the airway. Any time that arrives with the patient, such as the start of symptoms, the last time anyone saw the patient well or the last meal, belongs on paper before you leave the bedside. An unexplained patient who has become quiet is not a settled patient, so keep rechecking, keep them where you can see them, and ask for help while the patient can still talk to you.

Review Questions

1. A patient is newly confused, sweaty and cool, and your role allows you to check blood glucose. Why is the glucose checked early?

   A) The reading decides whether you call for help.

   B) Low blood glucose is a common cause that can be reversed within minutes.

   C) Confusion is always caused by a glucose problem.

   D) The airway cannot be assessed until the glucose is known.

2 A man with diabetes is shaky and sweating. He is awake, sits up and swallows normally, and his reading is low. Which action fits Entry 5.1?

- **A)** Lay him flat and give nothing by mouth.
- **B)** Wait for the ambulance before doing anything.
- **C)** Give plain water and recheck in an hour.
- **D)** Give the fast-acting sugar your protocol names, then recheck the reading after about 15 minutes.

3 A woman wakes with a drooping face and a weak left arm. Which time matters most to the receiving team?

- **A)** The last time she was known to be well.
- **B)** The time you arrived.
- **C)** The time she told someone.
- **D)** The time the call was connected.

4 Why should a responder not give aspirin to a patient with a suspected stroke unless medical direction says so?

- **A)** It works too slowly to help.
- **B)** It hides the facial droop.
- **C)** Some strokes are caused by bleeding, and aspirin could make bleeding worse.
- **D)** It is used only for chest pain.

5 A woman has a generalized seizure on a gym floor. Which action is correct?

- **A)** Hold her arms and legs still.
- **B)** Cushion her head, move hard objects away and note the time.
- **C)** Place a folded cloth between her teeth.
- **D)** Pour water on her face.

6 Which patient should be treated as possible sepsis?

- **A)** A patient with a mild cough, normal breathing and normal alertness.
- **B)** A patient with a sprained wrist and a fast heart rate from pain.
- **C)** A patient with a headache that eased after a drink of water.
- **D)** A patient with a skin infection who is newly drowsy, breathing fast, with cool, mottled skin.

7 An adult is found unresponsive with slow, shallow breathing and very small pupils. Which first step is correct?

- **A)** Make the scene safe, call emergency services, support breathing and, if your role allows and a device is available, use naloxone as directed.
- **B)** Make the patient vomit.
- **C)** Give a drink of water to wake the patient.
- **D)** Let the patient sleep and recheck in an hour.

8 Short case. A 70-year-old woman has had severe belly pain for 3 hours. She lies very still, her abdomen is rigid, and her skin is cool and pale with a fast pulse. List, in order, the first four things you would do.